

CLINICAL FINDINGS

Primary Syphilis

CUTANEOUS FINDINGS

Studies from the pre-antibiotic era typically found that 50 percent to 75 percent of persons who come in contact with a lesion of primary or secondary syphilis become infected. Even in the absence of clinical or serologic evidence of syphilis at the time of evaluation, 30 percent of exposed partners will develop the infection.

At the site of treponemal penetration, after an incubation period ranging from 10 to 90 days (average, 3 weeks), a dusky red macule appears. This lesion grows into a papule and then ulcerates to form a chancre. The length of the incubation period varies inversely with the number of inoculated *Treponema pallidum* organisms.

The chancre is typically round or oval, approximately 1 cm in diameter, and has sharply demarcated, regular, raised, firm, rubbery borders. The base is smooth and ham-colored, may appear clean or covered with yellowish crust or grayish slough, and is usually nontender in the absence of trauma or secondary infection.

Untreated, the chancre persists for 1 to 6 weeks and resolves within 1 to 2 weeks after treatment, healing without scarring. However, the classic "Hunter" chancre—characterized by raised, indurated borders and a slightly hemorrhagic necrotic base—is seen in only about 60 percent of cases. Deviations from this classic presentation are common, and in many instances, the lesions go unnoticed.

Multiple chancres may be present in up to 47 percent of cases and are often associated with condom use, which can shift inoculation sites. Other

complications include edema, phimosis, erosive balanitis, lymphangitis, and thrombophlebitis of the dorsal vein.

In industrialized countries, coinfection with genital herpes is the most likely explanation for multiple ulcers.

Common Sites of Involvement

- **In men:** glans penis, coronal sulcus, foreskin, and urethral meatus.
- Chancres at the urethral orifice may cause inflammatory phimosis, potentially leading to penile gangrene.
- A chancre on the inner surface of the foreskin may cause a brisk retraction when the foreskin is pulled back—a sign known as the **Dory flop**.
- **In women:** labia, fourchette, urethra, and perineum (in descending order of frequency).
- Chancres in women often exhibit edematous induration rather than the firm, cartilaginous consistency seen in men.